

This project demonstrates the implementation of a secure and cost-effective Azure cloud environment for CTU Training Solutions. The environment includes networking, virtual machines, identity management, security controls, monitoring and cost management.

This project improved my understanding of Azure cloud administration, infrastructure management, security implementation and cloud governance.

Architecture Diagram

AZURE ENVIRONMENT

Resource Group: NET731-RG

Virtual Network: VNet-NET731 (10.0.0.0/16)

Subnet: default-subnet (10.0.0.0/24)

NSG: NSG-NET731 (attached to subnet)

Virtual Machine: VM-NET731

Connected to subnet

Has Public IP

Storage Account: stnet731storage

Container 1: internal-docs (Private access)

Container 2: public-assets (SAS access)

Monitoring:

Azure Monitor enabled

Log Analytics Workspace (LAW-NET731)

Identity (Entra ID):

user-ops (Contributor)

user-audit (Reader)

user-data (Storage Blob Contributor)

SECURITY SECTION

RBAC

- user-ops → Contributor
 - user-audit → Reader
 - user-data → Storage Blob Data Contributor
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NSG Rules

- SSH (22) or RDP (3389) → My IP → Allow
 - HTTP (80) → Any → Allow
 - Deny All inbound → Default security
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Storage Security

- Private containers enabled
- SAS token used for limited access
- No public access allowed

MONITORING

- CPU monitored via Azure Monitor
- Network traffic tracked (in/out)
- Disk read/write monitored
- Logs stored in Log Analytics Workspace
- Alerts configured

COST SECTION

- Minimal cost (student/free tier)
- Main cost: Virtual Machine runtime
- VM stopped/deallocated when not in use
- No forecast due to limited usage history